

Project Check-In 2

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GitHub repo: <https://github.com/nurjahan-shiah/StudySync>

The delta of progress between this and the previous check-in:

Between Check-In 1 and Check-In 2, we moved from standing up the basic skeleton of StudySync to maturing the platform. In the first check-in, our work was mostly scaffolding and single-feature stubs e.g. the Docker Compose microservices setup, a login page, backend-only session creation, notification and announcement basics, and resource download/preview. By the second check-in, we'd pushed well past that: we hardened the entire auth and security layer (RBAC, JWT gateway validation, spoofed-header protection, session refresh/renewal), replaced our placeholder recommendation logic with a real scikit-learn model, and shipped a brand-new tasks microservice with full CRUD, assignment, and notification preferences from scratch.

We also rounded out features that were only half-built before, such as session scheduling, which became full-stack with conflict detection and cancellation; RSVP went from a backend endpoint to a complete discovery/participation UI; and we added things that weren't planned at all in CI1, like an AI-powered session notes summarizer and a leader-facing analytics dashboard. Alongside the new features, we put real effort into fixing bugs we'd introduced earlier (a silently unauthenticated admin dashboard, a non-functional ETL pipeline, a missing DB column) and expanded our curl test coverage significantly. Overall, CI1 was about getting things running; CI2 shows us making things correct, secure, and closer to production-ready.

Contributor	User Story	Task
Nurjahan Ahmed Shiah	US-D.1 US-F.1 US-F.3 US-F.4	> US-D.1 (Supabase File Upload Pipeline): Implemented direct browser-to-Supabase-Storage uploads with real-time progress tracking, bypassing the backend so large files don't hit the API gateway. Files are uploaded and namespaced by group and timestamp to prevent collisions, then registered as resource metadata via the resources service so they immediately appear in the group's resource list. > US-F.1 (Admin Dashboard): Extended the admin service with a full moderation layer; added endpoints to list all groups platform-wide (not just public ones). Also found and fixed a bug where the admin frontend was reading the JWT from the wrong localStorage key, meaning every admin API request was silently unauthenticated. > US-F.3 (ETL Pipeline & Delta Lake Integration): Fixed the ETL script, which had a function defined but never invoked, meaning the container ran and exited without writing any data. Also corrected the write mode from append to overwrite for the activity and resource tables so repeated runs don't duplicate rows in Delta Lake. > US-F.4 (Recommendation Engine): Replaced the placeholder overlap-counting logic with a real scikit-learn model; course enrollments and group-course associations are encoded as multi-hot vectors, and cosine similarity is computed between each user and group to produce ranked match scores. Also fixed a bug where writing recommendations via pandas' <code>to_sql</code> bypassed the ORM's default value generation, causing NOT NULL constraint violations on insert.
Uzma Alam	US-C.1 US-C.4	> Implemented a complete session scheduling system for the platform across both backend and frontend. For US-C.1, build the endpoint in the session microservice allowing

	US-G.2	group leaders to create study sessions with full field support. The endpoint validates that the scheduled time is in the future, and detects scheduling conflicts within the same group to prevent double-booking. The corresponding frontend component was implemented in the sessions page with a leader role check allowing group leaders to schedule sessions directly from the UI. > For US-C.4, implemented the session editing and cancellation functionality. The endpoint allows session creators or admins to update session details with a guard to prevent edits to already cancelled sessions. The EditSessionModal component was added to the session detail page with pre-filled fields, and a Cancel Session button triggers a confirmation dialog before cancelling. The is_cancelled field was also added to the StudySession model in shared_models.py. > For US-G.2, implemented an AI-powered session notes summarizer which extends the session detail page. The endpoint accepts rough session notes and calls the Groq API using the llama-3.3-70b-versatile model to generate a structured summary with a brief overview, key points, and action items. The SummariseModal frontend component allows the user to paste their notes and receive their AI-generated summary without leaving the page.
Ahmed (Saaram) Mustafa	M3, M4 (US-E.3 , US-E.5)	> Task Assigning & Tracking (US-E.3): Built a new tasks-service (port 8011) with a Task model and endpoints to assign, list, edit, delete, and update the status of tasks - with a title, description, assignee, due date, priority (Low/Medium/High), and status (To Do/In Progress/Completed). Leader-only assignment (validated against group membership), assignee/leader status updates, and an automatic in-app notification to the assignee on assignment. Wired into the API Gateway and Docker Compose. > Task frontend: Added a group-page Tasks tab where leaders assign, edit, and delete tasks and members update their own, plus a personal My tasks page listing tasks across all groups with one-click "Mark as Completed". Added due-date highlighting - red for overdue, yellow for deadlines within the week. > Notification Preferences (US-E.5): Added a NotificationPreference model and GET/PATCH /notification-preferences/{userId} endpoints (users edit only their own), with a preferences page of category toggles (sessions, announcements, tasks, resources, group activity) and a save confirmation. Enforced the preferences inside the shared notification helpers so disabled categories are never created - not just hidden - while the underlying task/announcement still occurs. > Fixes: Coerced date-only due dates to fix a task-creation validation error; made task assignment always notify the assignee; added the missing study_sessions.is_cancelled column to fix a session-creation crash; and made expired logins redirect to the login page instead of showing empty pages (with a longer token lifetime). > Testing: Wrote curl integration suites for both features - tests/curl/tasks (17 checks) and tests/curl/preferences (14 checks) - covering assignment, notifications, permissions, preference persistence, and enforcement.

Muhammad Fahad Sohail	US-C.3 / US-C.5 US-D.5	> Session Management & RSVP System (US-C.3 / US-C.5): Implemented the complete session discovery and participation workflow for study groups through the /sessions and /sessions/[id] pages. Students can browse study sessions organized into Upcoming and Past sections, with search and group-based filtering to quickly find relevant events. The session detail page displays session information, group context, and a live attendee list. Members can respond with Going, Maybe, or Not Going, with RSVP selections persisted and reflected immediately in the attendee list. The feature integrates with live data from the sessions-service backend to ensure session details and attendee responses remain accurate and up to date. > Upload Activity & Storage Stats (US-D.5): Implemented a leader-only analytics dashboard on the /resources page that provides visibility into resource usage and upload activity across groups managed by the current user. The widget displays key metrics including total files, recent uploads, file type distribution (PDFs, Images, and Other), and per-group file counts, helping leaders monitor library growth and engagement. It also includes a recent uploads section and group-level breakdowns using live resource data already retrieved from existing groups and resources endpoints. All statistics are computed client-side without requiring additional API requests.
Nusayba Hossain	US-A.2, US-A.3, US-A.4, US-D.4	> US-A.2 - Login and Session Lifecycle: Connected the Login page to the backend and completed token-based login, refresh-token handling, automatic session renewal, session persistence, and logout. > US-A.3 - Role-Based Access Control: Added centralized authorization for <i>student</i>, <i>group_leader</i>, and <i>admin</i> roles, with protected backend operations and correct 401 and 403 responses. > US-A.4 - API Gateway Authentication: Secured the API Gateway with JWT validation, protected private routes, forwarded trusted user identity headers, and blocked spoofed headers. > US-D.4 - Resource Deletion Permissions: Added permission-based resource deletion so only the uploader, relevant group leader, or admin can delete a resource. Added confirmation, loading, error handling, and automatic refresh.
Ali Shandhor	US-B.1, US-B.2, US-B.3	> US-B.1 - Group Discovery and Browse Page: Implemented the Study Groups browse page, allowing students to search for groups by name, filter them by course, and sort results by activity or member count. Connected the frontend to the backend so all group information is loaded dynamically. > US-B.2 - One-Click Join & Membership Management: Added one-click join and leave functionality, along with membership status indicators and automatic member count updates. Also prevented users from joining the same group more than once. > US-B.3 - Group Creation & Configuration Wizard: Implemented group creation, editing, and deletion for group leaders and administrators. Users can configure the group name, description, associated course, and visibility (public or private) through a simple create/edit interface, with full backend & frontend support for saving and updating group settings.